

How much storage does a renewable system need?

Storage and flexibility

Reading. Zerrahn et al. (2018), with the lecture note *Energy System Models* (ESM below), section 4, as the toolbox. Optional: Butters et al. (2025). Section, figure and equation numbers are the paper's own. Most questions are answered from the paper; those marked *Perspective* set it against ESM or against the evidence printed here.

Part 1 · Reading Zerrahn et al.

35 min

Why storage needs are two orders of magnitude below Sinn's

Zerrahn et al. (2018) reply to Sinn's claim that integrating wind and solar needs storage of the order of the surplus energy they produce. Sections 2–4 argue with data and a spreadsheet, section 5 with a cost-minimising model (part 2). This part reads sections 1–4 figure by figure, so keep the paper open at its figures. Their one analytical device is the *residual load duration curve* (§3.4), drawn in Figure 1 for Germany at 80% renewables. The storage has efficiencies η_c for charging and η_d for discharging, as in ESM.

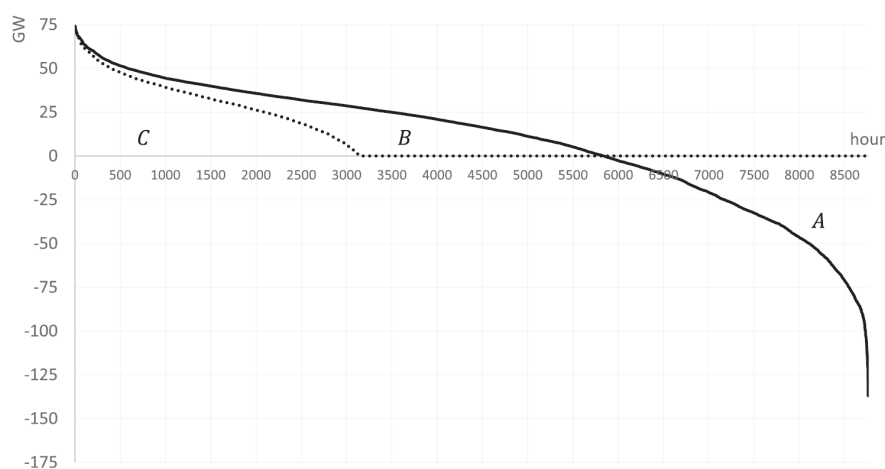


Figure 1: The paper's Fig. 4 (p. 265): Germany's residual load duration curve at 80% renewables, 2014 data. Solid: residual load; dotted: residual load after storage use, sorted again. Zerrahn et al. (2018), CC BY 4.0.

1. The claim (§1–2).

- (i) Explain what Sinn's original claim is.
- (ii) Which two assumptions do the authors argue Sinn makes in his analysis?
- (iii) What does Fig. 1 show about the claim, and where is Sinn in it?

2. Sinn's rule (§3). Explain Sinn's approach.

- (i) How does the paper simulate the storage requirement for a given share of renewables?
- (ii) How is the storage used: when does it charge, and when does it discharge?

- (iii) Do the results, generally, look realistic? Use Fig. 2 and Fig. 3.
Hint: set Fig. 2 against Germany's pumped hydro (§3.2), and compare the base years in Fig. 3.

3. The residual load duration curve (§3.4).

- (i) Explain the curve in Figure 1: what do the areas A, B and C measure?
 (ii) How does the residual load duration curve treat time, that is, the order of the hours?
Hint: compare one ten-hour surplus with ten one-hour surpluses, each followed by an hour of deficit. Area A is the same; is the tank?

4. Storage with power-oriented curtailment (§4.1).

- (i) Explain the basic idea using Fig. 5. What is the threshold, and what is area D?
 (ii) Explain Figs. 6 and 7. What do they tell us about the relationship between storage and curtailment?

5. Storage with energy-oriented curtailment (§4.2).

- (i) Explain the basic idea using Fig. 8. Where does it curtail, compared with Fig. 5?
 (ii) Explain Fig. 9 and compare it with Fig. 6.
Hint: read both figures at 50% renewables and 5% curtailment, and use question 3.

6. Perspective: perfect foresight versus heuristics. The storage of ESM §4.1 has perfect foresight: it holds energy for the best hour ahead. Sinn's heuristic releases it at the first positive residual load.

- (i) Into which hours of the residual load duration curve does each release? Footnote 16 says how Figure 1 comes to draw B as it does.
 (ii) What does foresight buy, and where does the paper put market reality (footnote 29)?

Stop point

The two assumptions; Sinn's rule stores every surplus and releases it at the first positive residual load; A passes through the tank and $\eta_c \eta_d$ of it comes back as B; the curve loses the order of the hours, and contiguous surplus runs set the tank. At 50% and 5% curtailment: 2.1 TWh becomes 0.3 TWh power-oriented, 19 GWh energy-oriented.

If you finish early

Three hours. Hour 1 is windy: load 3 GW, wind 3.5 GW. Hours 2 and 3 are calm with loads 1.5 and 2.5 GW; coal has 2 GW at 40 EUR/MWh and gas 2 GW at 80. A small battery (rating $\bar{s}$, two hours of tank, $\eta_c = 0.9$, $\eta_d = 1$) ends where it started. What does Sinn's rule earn per MW, what does the optimum earn, and what is θ_h at the optimum? Which of the paper's areas B_1 , B_2 , B_3 (§5.2) does the optimum reach that the rule does not?

The cost-minimising model

Section 5 of the paper replaces Sinn's arithmetic with a planner: a linear programme, equations (1)–(5), that chooses renewable, conventional and storage capacities together with their hourly dispatch, so as to serve a year of demand at least cost subject to a minimum renewable share. This part reads that section. The analysis is done with ESM's tools: multipliers on the constraints, and the first-order conditions they give. The problem, its Lagrangian and its conditions are printed below; your job is to say what they mean.

Notation. Upper-case letters are choices, everything else is data (p. 269). We keep the paper's i for annualised investment costs and v for variable costs, and its capacities: N^{se} for storage energy (MWh), N^{sp} for storage power (MW, the same for charging and discharging), N^c for conventional plant $c \in \{\text{base, peak}\}$ and N^r for renewables. Hourly, G_h^c and G_h^r are conventional and renewable generation, S_h^{in} and S_h^{out} the storage's charging and discharging, S_h its level and C_h curtailment; d_h is demand and γ_h the renewable capacity factor. We write η_c and η_d for the efficiencies and α for the minimum renewable share (the paper's φ).

The problem. For every hour h of the year, with the storage level equal in the first and the last hour, written as in ESM: every bounded variable between its bounds, and each constraint's multipliers on the right, underlined for a lower bound and overlined for an upper one.

$$\begin{aligned}
\min \quad & \underbrace{i^{se}N^{se} + i^{sp}N^{sp}}_{\text{storage energy and power}} + \underbrace{\sum_c i^c N^c + i^r N^r}_{\text{plant and renewables}} \\
& + \underbrace{\sum_{c,h} v^c G_h^c + \sum_h v^s (S_h^{in} + S_h^{out})}_{\text{variable costs}} \quad (1) \text{ total cost} \\
\text{s.t.} \quad & d_h = \sum_c G_h^c + G_h^r + S_h^{out} \quad (2) \text{ hourly balance} \quad (\lambda_h) \\
& 0 \leq G_h^c \leq N^c \quad (3a) \text{ plant capacity} \quad (\underline{\mu}_{-c,h}, \overline{\mu}_{c,h}) \\
& \gamma_h N^r = G_h^r + S_h^{in} + C_h \quad (3b) \text{ renewable energy} \quad (v_h) \\
& G_h^r \geq 0, \quad C_h \geq 0 \quad \text{renewable use} \quad (\underline{\mu}_{-r,h}, \underline{\omega}_h) \\
& S_h = S_{h-1} + \eta_c S_h^{in} - S_h^{out} / \eta_d \quad (4a) \text{ storage balance} \quad (\theta_h) \\
& 0 \leq S_h^{in} \leq N^{sp} \quad (4b) \text{ charging rating} \quad (\underline{\phi}_h^c, \overline{\phi}_h^c) \\
& 0 \leq S_h^{out} \leq N^{sp} \quad (4c) \text{ discharging rating} \quad (\underline{\phi}_h^d, \overline{\phi}_h^d) \\
& 0 \leq S_h \leq N^{se} \quad (4d) \text{ energy capacity} \quad (\underline{\psi}_h, \overline{\psi}_h) \\
& \sum_{c,h} G_h^c \leq (1 - \alpha) \sum_h d_h \quad (5) \text{ renewable share} \quad (\sigma) \\
& N^{se}, N^{sp}, N^c, N^r \geq 0 \quad \text{capacities} \quad (\underline{\beta}^{se}, \underline{\beta}^{sp}, \underline{\beta}^c, \underline{\beta}^r)
\end{aligned}$$

The Lagrangian. As in ESM §4.1, the planner maximises minus the cost, and each constraint enters as a slack that is nonnegative when the constraint holds, multiplied by its multiplier; a lower bound $x \geq 0$ enters as $\underline{\mu} x$. The multipliers have ESM's names where ESM has the constraint, and the paper's two further variables get $\underline{\mu}_{-r,h}$ and $\underline{\omega}_h$ for their lower bounds and $\underline{\beta}$ for the capacities, as in ESM's greenfield model. The multipliers on the equalities, λ_h , v_h and

θ_h , are free; those on the inequalities are nonnegative.

$$\begin{aligned}
L = & \underbrace{- \left(i^{se} N^{se} + i^{sp} N^{sp} + \sum_c i^c N^c + i^r N^r + \sum_{c,h} v^c G_h^c + \sum_h v^s (S_h^{in} + S_h^{out}) \right)}_{\text{minus the total cost (1)}} \\
& + \underbrace{\sum_h \lambda_h \left(\sum_c G_h^c + G_h^r + S_h^{out} - d_h \right)}_{\text{(2) hourly balance}} + \underbrace{\sum_{c,h} \left(\bar{\mu}_{c,h} (N^c - G_h^c) + \underline{\mu}_{-c,h} G_h^c \right)}_{\text{(3a) plant capacity}} \\
& + \underbrace{\sum_h v_h (\gamma_h N^r - G_h^r - S_h^{in} - C_h)}_{\text{(3b) renewable energy}} + \underbrace{\sum_h \left(\underline{\mu}_{-r,h} G_h^r + \underline{\omega}_h C_h \right)}_{\text{renewable use}} \\
& + \underbrace{\sum_h \theta_h (S_{h-1} + \eta_c S_h^{in} - S_h^{out} / \eta_d - S_h)}_{\text{(4a) storage balance}} + \underbrace{\sum_h \left(\bar{\phi}_h^c (N^{sp} - S_h^{in}) + \underline{\phi}_h^c S_h^{in} \right)}_{\text{(4b) charging rating}} \\
& + \underbrace{\sum_h \left(\bar{\phi}_h^d (N^{sp} - S_h^{out}) + \underline{\phi}_h^d S_h^{out} \right)}_{\text{(4c) discharging rating}} + \underbrace{\sum_h \left(\bar{\psi}_h (N^{se} - S_h) + \underline{\psi}_h S_h \right)}_{\text{(4d) energy capacity}} \\
& + \underbrace{\sigma \left((1 - \alpha) \sum_h d_h - \sum_{c,h} G_h^c \right)}_{\text{(5) renewable share}} + \underbrace{\beta^{se} N^{se} + \beta^{sp} N^{sp} + \sum_c \beta^c N^c + \beta^r N^r}_{\text{capacities}}
\end{aligned}$$

The first-order conditions. Differentiate L with respect to each variable and set the derivative to zero. Complementary slackness holds on every bound: a bound's multiplier is zero unless the bound binds.

Capacities. One condition for each capacity: minus its annuity, the multipliers of the constraints that capacity appears in, summed over the year, and the multiplier on its own lower bound.

$$\begin{aligned}
N^{sp} : & -i^{sp} + \sum_h \left(\underbrace{\bar{\phi}_h^c}_{(4b)} + \underbrace{\bar{\phi}_h^d}_{(4c)} \right) + \underbrace{\beta^{sp}}_{N^{sp} \geq 0} = 0 & N^{se} : & -i^{se} + \sum_h \underbrace{\bar{\psi}_h}_{(4d)} + \underbrace{\beta^{se}}_{N^{se} \geq 0} = 0 \\
N^c : & -i^c + \sum_h \underbrace{\bar{\mu}_{c,h}}_{(3a)} + \underbrace{\beta^c}_{N^c \geq 0} = 0 & N^r : & -i^r + \sum_h \gamma_h \underbrace{v_h}_{(3b)} + \underbrace{\beta^r}_{N^r \geq 0} = 0
\end{aligned}$$

Generation. One condition for each use of an hour's energy: the price λ_h against what conventional and renewable output cost at the margin, and the value v_h of renewable energy against curtailment.

$$\begin{aligned}
G_h^c : & -v^c + \lambda_h - \underbrace{\sigma - \bar{\mu}_{c,h} + \underline{\mu}_{-c,h}}_{(3a)} = 0 & G_h^r : & \lambda_h - v_h + \underbrace{\underline{\mu}_{-r,h}}_{G_h^r \geq 0} = 0 \\
C_h : & -v_h + \underbrace{\underline{\omega}_h}_{C_h \geq 0} = 0
\end{aligned}$$

Storage operation. One condition for each storage variable, with θ_h , the multiplier on the storage balance (4a), the value of a MWh in the tank at the end of hour h . In the last hour the cyclic condition takes the place of θ_{h+1} .

$$\begin{aligned}
 S_h^{in} : \quad & -v^s - v_h + \eta_c \theta_h - \underbrace{\bar{\phi}_h^c + \phi_h^c}_{(4b)} = 0 & S_h^{out} : \quad & -v^s + \lambda_h - \theta_h / \eta_d - \underbrace{\bar{\phi}_h^d + \phi_h^d}_{(4c)} = 0 \\
 S_h : \quad & -\theta_h + \theta_{h+1} - \underbrace{\bar{\psi}_h + \psi_h}_{(4d)} = 0
 \end{aligned}$$

1. The planner's problem (§5.1).

- (i) Explain the planner's problem in your own words: what is minimised, what is chosen, and what each of (2)–(5) requires. What does each multiplier price, and why must those on the inequalities be nonnegative?
- (ii) Explain the four capacity conditions and the condition for G_h^c in words: which hours pay for a MW of pumps, which for a MWh of basin, which for a MW of turbines, and what σ does to a conventional plant. Why does the planner build turbines whose output is sometimes worth nothing?
Hint: use the conditions for G_h^r , S_h^{in} and C_h to say what v_h is in an hour when renewables serve load, when they fill the tank, and when they are curtailed.
- (iii) Put the condition for N^{se} together with the surplus of part 1, which arrives in a few long runs. Why does the planner stop at a tank far below Sinn's, and why is it content to curtail?
- (iv) *Perspective:* the remaining conditions, for S_h^{in} , S_h^{out} and S_h , are those of the storage model of ESM §4.1. How does the planner's problem differ from that model, and which of ESM's results carry over unchanged?

2. The residual load duration curve at the optimum (§5.2). The paper's Fig. 10, at 80%.

- (i) In the paper's words, what are the areas D_1 , D_2 , B_1 , B_2 and B_3 ?
- (ii) What are the two "economic values of electrical storage" the paper names, and which areas carry each?

3. The numbers (§5.3).

- (i) What does the paper find for Germany at 50% and at 90%: storage energy, storage power, the energy-to-power ratio, and what does it conclude about long-term storage?
- (ii) Why does the energy-to-power ratio rise with the renewable share, and why does optimal curtailment rise with it?
Hint: which hours pay for a MW of pumps, and which for a MWh of basin? How does each set of hours change as the fleet grows?

4. Power-to-x (§5.4). The paper's last experiment adds a flexible new load.

- (i) What is added to the model, and what happens to storage and curtailment at 50% and at 90% (Fig. 12)?
- (ii) In one sentence: what does a shiftable load share with a storage, and what does it lack?

Stop point

The first-order conditions in words: a MW of pumps is paid for by the hours it runs at its rating, a MWh of basin by the hours the tank is full, a MW of turbines by the hours its output is not curtailed, and σ is a tax on conventional energy. A skewed surplus fills the last MWh of basin too rarely to pay for it. Germany: 35 GWh and 6 GW at 50%, no long-term storage, and 35 to 4 GWh with power-to-x.

If you finish early

Appendix C re-runs the model with lithium-ion costs: cheap in MW, costly in MWh, and a higher round-trip efficiency. Use the first-order conditions to predict what happens to N^{se} , N^{sp} , the energy-to-power ratio and curtailment, then check against the appendix. What does the change say about the paper's §6 point that storage is a family of technologies rather than one?

What the paper's model says about Denmark

The paper runs its planner on Germany's 2014 data. We have run the same model, equation for equation, on DK1's load and weather for 2024 (24.1 TWh of demand, a peak of 4.2 GW), with the renewable share target swept from 50 to 95%. Table 1 reports the solutions, Figure 2 places them on the axes of the paper's Fig. 1, and Figure 3 draws the 80% solution as a residual load duration curve, the paper's Fig. 10 for DK1. From part 2 you need one result: σ , the multiplier on the share constraint, enters a conventional plant's condition as a tax of σ per MWh of conventional energy.

How the model is set up. Costs are ESM's tables for 2050, annuitised. There is one renewable technology, in DK1's own mix of 48% onshore wind, 21% offshore wind and 31% solar, with a capacity factor of 28% and an annuity of 89 thousand EUR per MW and year. Two gas plants play the paper's base and peak roles: a combined-cycle turbine (CCGT, annuity 100 thousand EUR per MW, 63 EUR/MWh to run) and an open-cycle one (OCGT, 45 thousand, 86 EUR/MWh). Storage is ESM's lithium-ion battery with power and energy priced separately, 9.7 thousand EUR per MW of inverter and 6.4 thousand per MWh of cells a year, which is the paper's Appendix C case. As in the paper, curtailment is free, the storage charges only from renewables, there is no carbon price, and the system is an island.

Three variations. The lower blocks of Table 1 change one thing each. *With imports* adds ESM's 5.5 GW import connection at the neighbours' 2024 hourly prices; imports count as conventional energy in the share constraint (5). *Long* prices the cells at a tenth of their cost, a stand-in for a long-duration technology. *Free* lets the storage charge from any source, not only from renewables. Question 1 uses the second, question 4 the first and the third.

1. DK1 against the paper's conclusions.

- (i) Read DK1's storage at 80% from Table 1. Express it in percent of annual demand and in hours of DK1's average load, and place it on Figure 2(a) next to Germany (§5.3) and next to Sinn's numbers (§2).
- (ii) Set DK1's curtailment at 80% against Germany's optimal curtailment in §5.3.
- (iii) The "long" rows price the cells at a tenth of their cost. What energy-to-power ratio does the planner then choose, and is that the "true long-term storage" §5.3 says is never needed?

target	achieved	N^r , GW	conv., GW	N^{se} , GWh	N^{sp} , GW	E/P, h	curtailed %	σ , €/MWh
<i>The paper's model on DK1, islanded</i>								
50%	68%	7.3	3.1	3.91	0.48	8.1	7.5	0.0
70%	70%	7.6	3.0	5.15	0.60	8.6	8.8	5.1
80%	80%	9.4	2.7	15.97	1.43	11.2	14.9	41.9
90%	90%	12.1	2.2	37.77	2.36	16.0	25.6	145.2
95%	95%	13.0	1.8	86.93	2.79	31.2	26.6	458.5
<i>With the 5.5 GW import backstop, imports counted as conventional energy</i>								
50%	61%	6.5	0.0	10.64	2.12	5.0	4.1	0.0
70%	70%	7.6	0.0	12.07	2.20	5.5	7.2	13.4
80%	80%	9.2	0.0	20.33	2.47	8.2	13.1	51.0
90%	90%	12.0	0.0	39.96	2.84	14.1	25.0	167.9
95%	95%	13.0	0.0	87.34	2.99	29.2	26.6	477.2
<i>Two sensitivities, islanded: cells at a tenth of the cost (long), and charging from any source (free)</i>								
80%, long	94%	10.0	1.5	189.01	2.77	68.2	5.1	0.0
90%, long	94%	10.0	1.5	189.01	2.77	68.2	5.1	0.0
80%, free	80%	9.4	2.7	15.97	1.43	11.2	14.9	41.9
90%, free	90%	12.1	2.2	37.77	2.36	16.0	25.6	145.2

Table 1: The paper's model on DK1. "Achieved" is the renewable share the solution actually has; σ is the multiplier on the share constraint (5), in EUR per MWh of conventional energy; curtailment is a share of the renewable energy the weather offered.

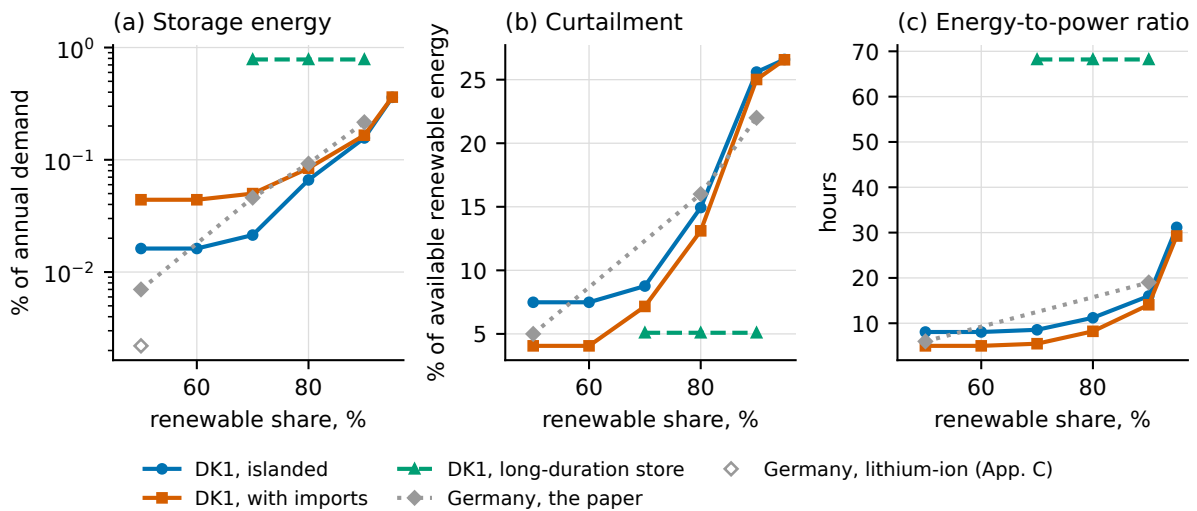


Figure 2: DK1 against the paper's Germany: (a) storage energy as a share of annual demand, the axis of the paper's Fig. 1; (b) curtailment as a share of the renewable energy available; (c) the energy-to-power ratio of the storage. Germany's points are the paper's numbers.

2. The residual load duration curve for DK1.

- The dashed curve in Figure 3 has three flat parts. Explain what sets each. The lowest is minus the storage's power rating: check it against the N^{sp} column of Table 1 at 80%.
- The figure is the paper's Fig. 10 for DK1 (§5.2, part 2's question 2). Mark the five areas D_1 , D_2 , B_1 , B_2 and B_3 on it.

3. What the target costs.

- σ is zero at a 50% target. What does that mean, and what share does the planner reach on cost alone? Use the costs in the setup to say why it is there.
Hint: divide the renewable annuity by the MWh a MW delivers in a year.

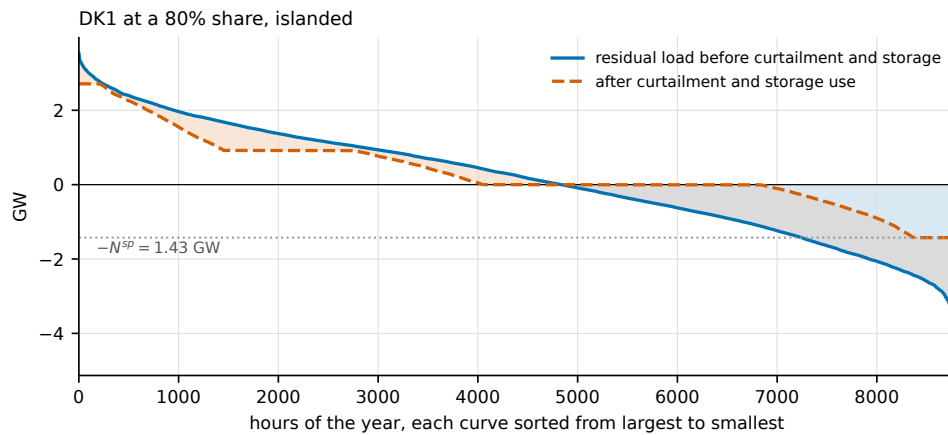


Figure 3: DK1's residual load duration curve at an 80% share, islanded, drawn as the paper's Fig. 10: before curtailment and storage, and after both, each sorted on its own. The dotted line is minus the storage's power rating.

- (ii) Read σ at 80% from Table 1 and convert it to a carbon price, using that a CCGT burns 1/0.6 MWh of gas per MWh with 0.198 t of CO_2 per MWh of gas.
- 4. *Perspective: \$6 tested on DK1.* Table 1 tests two things the paper left out, and in both DK1 disagrees with it: interconnection (\$6, fifth point), which the paper expects to lower storage needs, and the remark at the end of §5.2 that charging from conventional plants “would strengthen both economic values”.
 - (i) With imports, storage *rises*: 20.33 GWh against 15.97 at 80%. Footnote 17 says what interconnection does to the surplus. What can a one-way import connection do for area A, and what can it not?
 - (ii) The “free” rows are identical to the islanded ones. Why does charging from the gas plants never pay here? The two plants in the setup above, their running costs, and σ are what you need.

Stop point

DK1's storage at 80% in GWh, in percent of annual demand and in hours of average load, against Germany's; the share the planner reaches unforced; and σ at 80% as a carbon price.

If you finish early

Apply Sinn's 3.23% at 89% to DK1 and set it against the table's 90% row. Then Appendix B: the weather year changes the optimal storage by a factor of five. Name the statistic of the hourly residual load that would predict which of DK1's years 2019 to 2024 needs the most storage, and say why the residual load duration curve cannot provide it.